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Michael Curry

SUMMARY

A seasoned **Senior Data Engineer** over 10+ years of extensive experience, I specialize in designing, building, and optimizing data pipelines and architectures on **Azure** and **AWS** platforms. Proficient in **ETL/ELT processes**, data integration, and big data technologies. I deliver scalable, secure, and high-performance solutions tailored to meet enterprise requirements. I excel at transforming raw data into actionable insights, enabling data-driven decision-making and supporting business growth. My expertise extends to modernizing legacy systems, implementing end-to-end data platforms, and enabling real-time analytics for mission-critical applications.

Core Competencies

- Proficient in **Data Azure factory, Databricks, Synapse Analytics, Data Lake** and **AWS Glue, S3, Redshift, Athena, Lambda, EC2**.
- Expertise in designing and managing **ETL/ELT pipelines**, **data lakes** and **data warehouses** using tools like **Python**, **Apache Spark, PySpark**, and **SQL**.
- Hands-on experience with **Hadoop, Hive, Kafka**, and **Spark Streaming**, ensuring efficient processing of large-scale data.
- Built real-time data ingestion pipelines using **Kafka, Kinesis**, and **Azure Event Hubs**, enabling real-time analytics.
- Extensive experience with **relational databases** (MySQL, PostgreSQL, SQL Server) and **NoSQL databases** (MongoDB, DynamoDB, Cassandra, Cosmos DB).
- Created interactive dashboards using **Power BI, Tableau**, and **QuickSight**, offering actionable insights for stakeholders.
- Designed workflows using **Apache Airflow**, **Azure Data Factory**, and **AWS Step Functions** for seamless automation and orchestration.
- Implemented **CI/CD pipelines** with tools like **Jenkins, GitHub Actions** and **Azure DevOps** for streamlined data workflow deployment.
- Ensure adherence to **data governance**, encryption standards and compliance with frameworks like **GDPR, HIPAA** and **SOC 2**

EDUCATION

Master of Science (MS) in Computer Science – Santa Clara University

June 2013 - April 2015, Santa Clara, CA

Bachelor of Science (BS) in Computer Science – Santa Clara University

September 2009 - May 2013, Santa Clara, CA

EXPERIENCE

JP MorganChase | New York, NY — Senior Data Engineer

July 2023 - PRESENT

As a Senior Data Engineer, I designed and implemented advanced data solutions for Capital Market clients. The project cantered on modernizing and optimizing trading platforms to address the complexities of capital markets. By leveraging Azure Data Services, I developed scalable, secure, and high-performance data pipelines to process trading activities efficiently, ensuring compliance with financial regulations.

Key focus areas included:

- Real-Time Decision-Making: Implemented real-time analytics to provide faster insights into trading activities and market trends.
- Data Governance and Security: Integrated robust data governance frameworks and security measures to meet stringent financial industry standards.
- Scalability and Optimization: Enhanced data processing pipelines for improved performance and operational efficiency.

Key Outcomes

- Delivered a unified data platform that improved data accessibility, enabled faster decision-making, and supported advanced analytics for Capital Market operations.
- Reduced data processing time by 40% through optimized pipelines and cloud-native technologies.
- Improved data governance and security compliance, ensuring adherence to financial regulations.
- Enabled stakeholders to gain actionable insights through real-time dashboards and predictive analytics.
- Designed and implemented end-to-end ETL pipelines using Azure Data Factory to ingest, process, and transform trading data efficiently. Used Python for automation.
- Leveraged Azure Databricks and Apache Spark for large-scale data processing and real-time analytics on trading and market datasets.
- Managed seamless data integration across various sources, including on-premises systems using Azure Datawarehouse, Azure Data Lake and APIs, ensuring smooth and reliable workflows.
- Developed scalable and optimized data models for trading data using star and snowflake schema principles.
- Designed and maintained Azure Synapse Analytics for high-performance data warehousing, supporting fast querying and reporting.
- Optimized data structures and queries for quick retrieval of trading insights, improving decision-making efficiency.
- Implemented robust data governance policies, including data lineage, metadata management, and auditing, ensuring compliance with financial regulations.
- Secured data using role-based access control (RBAC), encryption, and monitoring tools such as Azure Security Center and Azure Key Vault.
- Created interactive dashboards and reports in Power BI to visualize trade performance, risk metrics, and compliance KPIs for stakeholders.
- Conducted advanced data analysis, integrating Azure Machine Learning models for predictive insights into trading trends and market forecasting.
- Built real-time streaming pipelines with Azure Event Hubs and Azure Stream Analytics for processing trading events and aggregating market data.
- Established alert mechanisms for anomalies in trading patterns to identify potential risks or opportunities proactively.
- Monitored and optimized system performance using Azure Monitor and Log Analytics, resolving bottlenecks to meet SLA requirements.
- Migrated on-premises databases and applications to Azure cloud infrastructure, ensuring minimal downtime and data integrity during the transition.
- Developed data archival strategies using Azure Blob Storage and Azure Data Lake for cost-effective long-term storage solutions.

US Bank | Mineapolis, MN — Data Engineer

Jan 2018 - Jun 2023

As a Data Engineer in an e-commerce and Retail Platform, the focus was on designing, developing, and optimizing data pipelines to ensure seamless data flow across multiple systems. This enabled real-time decision-making and advanced analytics, with the aim of enhancing customer experiences, improving inventory management, and streamlining supply chain operations through efficient and scalable data engineering solutions.

Key Focus and Objectives:

- Build a unified data platform to centralize e-commerce and retail data, providing insights into customer behaviour, inventory levels, sales trends, and operational efficiency.
- Enable predictive analytics and personalized customer experiences by integrating machine learning capabilities.
- Real-time inventory tracking to prevent stockouts and reduce overstocking, thereby minimizing operational costs.
- Empowering business leaders with real-time dashboards and predictive analytics for informed decision-making.

Key Outcomes

- Delivered personalized shopping recommendations and improved website performance through real-time data processing.
- Empowered business teams with actionable insights, enabling faster decision-making and optimized marketing campaigns.
- Streamlined inventory management, reducing operational costs and stock discrepancies by 20%.
- Built a robust and scalable data infrastructure that adheres to data privacy regulations, ensuring long-term usability and compliance.
- Enabled data-driven dynamic pricing strategies and targeted promotions, contributing to a measurable increase in revenue.
- Developed and optimized ETL/ELT pipelines to extract, transform, and load data from diverse sources such as transactional systems, APIs, and third-party platforms into a centralized data warehouse, ensuring seamless integration and data availability.
- Enabled real-time data ingestion for critical systems, including inventory management and sales dashboards, to support timely decision-making and operational efficiency.
- Orchestrated workflows using tools like Apache Airflow, Azure Data Factory, and AWS Glue, automating data pipelines and minimizing manual intervention using Python.
- Integrated data across e-commerce platforms (e.g., Shopify, Magento) and retail POS systems, establishing smooth and reliable data flows for inventory, sales, and customer behavior analysis.
- Collaborated with cross-functional teams to deliver real-time analytics for inventory management, sales performance, and customer insights.
- Designed scalable data models utilizing star and snowflake schemas to support analytics for sales, customer segmentation, inventory optimization, and supply chain operations.
- Created logical and physical data models to enhance data accessibility and improve system performance.
- Developed interactive dashboards in Power BI and Tableau, offering actionable insights into sales trends, customer segmentation, and marketing campaign success.
- Implemented real-time streaming pipelines with tools like Kafka and Azure Event Hubs, enabling inventory level monitoring and real-time alerts for low stock scenarios.
- Built performance metrics (KPIs) for sales, customer behavior, supply chain efficiency, and marketing effectiveness to measure and drive improvements.
- Optimized inventory management through data-driven algorithms, minimizing stockouts and overstock situations, improving cost efficiency.
- Enabled dynamic pricing models by leveraging sales trends, market conditions, and competitor data to maximize revenue and customer satisfaction.

Liberty Mutual | Boston, MA — Data Engineer

May 2015 - Dec 2017

- Designed and implemented Azure Data Factory (ADF) pipelines to integrate data from on-premise systems (MySQL, Cassandra) and cloud sources (Azure Blob Storage, Azure SQL DB). Applied transformations using Python and loaded the processed data into Azure Synapse Analytics for reporting and analysis.
- Architected and developed frameworks to ingest data in various formats (JSON, XML, CSV) from multiple sources into Azure SQL Data Warehouse, ensuring data accuracy and reliability.
- Configured and managed resources across clusters using Azure Kubernetes Service (AKS) to optimize performance and cost efficiency.
- Monitored Spark clusters using Log Analytics and Ambari Web UI, transitioning log storage from Cassandra to Azure SQL Data Warehouse, which improved query performance significantly.
- Built data ingestion pipelines on Azure HDInsight Spark clusters using ADF and Spark SQL.
- Created Databricks notebooks for data curation, batch, and streaming data processing, leveraging Spark Streaming to analyze real-time healthcare data such as patient admissions, claims processing, and medical device monitoring.
- Developed interactive dashboards and reports using Power BI and SQL Server Reporting Services (SSRS) to provide actionable insights into patient outcomes, claims status, and operational metrics for healthcare management.
- Designed department-specific dashboards with role-based access control using Azure Active Directory (AAD) to manage permissions and ensure compliance with healthcare data privacy regulations like HIPAA.
- Utilized Azure Logic Apps to automate workflows such as scheduling batch jobs, triggering email notifications, and integrating ADF pipelines with other Azure services for seamless operations.
- Developed frameworks with error logging in ADF v2 to ensure accurate and traceable data processing.
- Built real-time data streaming pipelines using Spark Streaming in Databricks to monitor critical healthcare metrics, such as patient discharge rates or appointment no-shows, and trigger alerts for anomalies.
- Enabled continuous data flow for healthcare devices and systems using tools like Azure Event Hubs.
- Performed data migration from on-premise databases to Azure Databricks and Azure SQL Data Warehouse, ensuring minimal downtime and maintaining data integrity.
- Migrated CRM data and enabled seamless integration for better patient relationship management and operational workflows.
- Identified and onboarded various healthcare data sources to support data science projects like predictive analytics for patient readmission and disease progression.
- Engaged stakeholders to increase visibility of analytics initiatives and delivered high-level and low-level designs (HLD/LLD) for technical and business solutions.
- Integrated AAD authentication with Cosmos DB to secure data requests and demonstrated security features to stakeholders.
- Implemented role-based access control (RBAC), encryption, and monitoring tools to ensure compliance with healthcare standards such as HIPAA.
- Worked on data integration and migration within CRM systems, ensuring seamless transfer of patient and provider data while maintaining integrity.
- Created Tableau visualizations to track healthcare KPIs, patient outcomes, and operational performance, optimizing dashboards for better usability and performance.